

Syed Huzafa Ali

23K-0004

BAI-7A

Agentic AI

Date

Assignment Alpha

Q1.

1). Bits per frame = 29491200 bits/frame

pixels per frame = $1280 \times 720 \Rightarrow 921600$ pixels

$X = \frac{29491200}{921600} \Rightarrow 32$ bits/pixel

2). Bytes per frame = $\frac{29491200}{8} \Rightarrow 3686400$ bytes/frame

3). $70 \times 1048576 \times 8 \Rightarrow 587202560$ bits/s

Q2. Q =	1.5	3	K =	1.1	2.2	V =	0.2	0.3
	2	4		1	3		2	0.3
	2.2	4.4		1	3		2.2	0.3
	2.4	4.4		1.1	3		2.2	0.3

$d_k = 2$

Attention = $\text{Softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V$

$QK^T =$	1.5	3	1.1	1	1	1.1
	2	4	2.2	3	3	3
	2.2	4.4				
	2.4	4.4				

$QK^T =$	8.25	10.5	10.5	10.65
	11	14	14	14.2
	12.1	15.4	15.4	15.62
	12.32	15.6	15.6	15.84

$QK^T/\sqrt{d_k} =$	5.834	7.425	7.425	7.531
	7.778	9.899	9.899	10.041
	8.556	10.889	10.889	11.045
	8.716	11.031	11.031	11.201

$\text{Softmax}(QK^T/\sqrt{d_k}) =$	0.0614	0.3016	0.3016	0.3353
	0.0366	0.3056	0.3056	0.3522
	0.0297	0.3062	0.3062	0.3579
	0.0301	0.3045	0.3045	0.3609

$\times$

Attention =	0.0614	0.3016	0.3016	0.3353	0.2	0.3
	0.0366	0.3056	0.3056	0.3522	2	0.3
	0.0297	0.3062	0.3062	0.3579	2.2	0.3
	0.0301	0.3045	0.3045	0.3609	2.2	0.3

Attention =	2.0167	0.3	→ AI
	2.0657	0.3	→ Governance
	2.0794	0.3	→ Emerging
	2.0789	0.3	→ Course

Q3.	Wind	Humidity	Rain	$P(\text{Yes}) = 0.6$ $P(\text{No}) = 0.4$
	Weak	High	Yes	
	Strong	High	Yes	
	Weak	Low	No	
	Weak	Low	Yes	
	Strong	Low	No	

1). $P(\text{feature}|\text{class}) = \frac{\text{count} + 1}{N + k} \quad \therefore k = 2$

Wind given Rain:

	Yes ($n = 3$)	No ($n = 2$)
$P(\text{Weak} \text{Yes})$	$\frac{2+1}{3+2} \Rightarrow 0.6$	$\frac{1+1}{2+2} \Rightarrow 0.5$
$P(\text{Strong} \text{c})$	$\frac{1+1}{3+2} \Rightarrow 0.4$	$\frac{1+1}{2+2} \Rightarrow 0.5$

Humidity given Rain:

	Yes ($n = 3$)	No ($n = 2$)
$P(\text{High} \text{c})$	$\frac{2+1}{3+2} \Rightarrow 0.6$	$\frac{0+1}{2+2} \Rightarrow 0.25$
$P(\text{Low} \text{c})$	$\frac{1+1}{3+2} \Rightarrow 0.4$	$\frac{2+1}{2+2} \Rightarrow 0.75$

We use smoothing, because $P(\text{High}|\text{No})$ would be '0' without it.

Wind + Humidity	score (Yes)	score (No)	$P(\text{Yes})$	$P(\text{No})$	Decision
Weak + High	$0.6 \Rightarrow 0.216$	$0.4 \times 0.5 \times 0.25 \Rightarrow 0.05$	0.812	0.188	Rain, No Delivery
Weak + Low	$0.6 \times 0.6 \times 0.4 \Rightarrow 0.144$	$0.5 \times 0.75 \times 0.4 \Rightarrow 0.15$	0.49	0.51	No Rain, Delivery
Strong + High	$0.6 \times 0.6 \times 0.4 \Rightarrow 0.144$	$0.4 \times 0.5 \times 0.25 \Rightarrow 0.05$	0.742	0.258	Rain, No Delivery
Strong + Low	$0.4 \times 0.4 \times 0.6 \Rightarrow 0.096$	$0.5 \times 0.75 \times 0.4 \Rightarrow 0.15$	0.39	0.61	No Rain, Delivery

$$2 \text{ Entropy} = -0.6 \log_2 0.6 - 0.4 \log_2 0.4 \Rightarrow 0.971$$

For Humidity:

①. High: 2 Yes, 0 No

$$H(\text{High}) = 0$$

②. Low: 1 Yes, 2 No

$$H(\text{Low}) = -\frac{1}{3} \log_2 \left(\frac{1}{3}\right) - \frac{2}{3} \log_2 \left(\frac{2}{3}\right)$$

$$H(\text{Low}) = 0.918$$

$$H(S|\text{Humidity}) = \frac{2}{5}(0) + \frac{3}{5}(0.918)$$

$$\Rightarrow 0.551$$

$$I.G(\text{Humidity}) = 0.971 - 0.551$$

$$\Rightarrow 0.42$$

For Wind:

①. Strong: 1 Yes, 1 No

$$H(\text{Strong}) = 1$$

②. Weak: 2 Yes, 1 No

$$H(\text{Weak}) = -\frac{2}{3} \log_2 \left(\frac{2}{3}\right) - \frac{1}{3} \log_2 \left(\frac{1}{3}\right)$$

$$H(\text{Weak}) = 0.918$$

$$H(S|\text{Wind}) = \frac{2}{5}(1) + \frac{3}{5}(0.918)$$

$$\Rightarrow 0.951$$

$$I.G(\text{Wind}) = 0.971 - 0.951$$

$$\Rightarrow 0.02$$

Root = Humidity.

Wind:

①. Weak: 1 Yes, 1 No

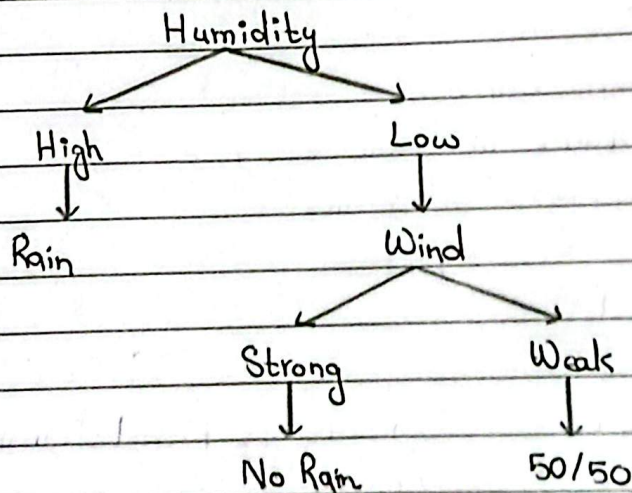
$$H(\text{Weak}) = 1$$

②. Strong: 0 Yes, 1 No

$$H(\text{Strong}) = 0$$

$$\frac{2}{3}(1) + \frac{1}{3}(0) \Rightarrow 0.667$$

$$I.G(\text{Wind}|\text{Low}) = 0.918 - 0.667 \Rightarrow 0.252$$



3. Wind (W): Strong = 1, Weak = 0, Humidity (H): High = 1, Weak = 0

Rain (R): Yes = 1, No = 0

Wind	Humidity	Rain	$z = w_1x_1 + w_2x_2 + b$
0	1	1	$\sigma(z) = \frac{1}{1+e^{-z}}$
1	1	1	
0	0	0	If $P \geq 0.5$ then Rain = Yes
0	0	1	Loss = $-\frac{1}{m} \sum [y_i \ln p_i + (1-y_i) \ln (1-p_i)]$
1	0	0	

Iteration 1: $\alpha = 0.1$, $w_1 = w_2 = b = 0$

W	H	z	P	$e = y - p$	Loss = $(-1/5)(-0.6931 \times 5)$
0	1	0	0.5	0.5	$\Rightarrow 0.6931$
1	1	0	0.5	0.5	$b_{new} = 0 + 0.1(0.5 + 0.5 - 0.5 + 0.5 - 0.5)$
0	0	0	0.5	-0.5	$b_{new} = 0.05$
0	0	0	0.5	0.5	$w_1 = 0.1(0.5 \times 1 + 1 \times -0.5) \Rightarrow 0$
1	0	0	0.5	-0.5	$w_2 = 0.1(0.5 + 0.5) \Rightarrow 0.1$

Iteration 2: $\alpha = 0.1$, $w_1 = 0$, $w_2 = 0.1$, $b = 0.05$

W	H	z	P	$e = y - p$	Loss = $(-1/5)(-3.3474)$
0	1	0.15	0.5374	0.4626	$\Rightarrow 0.6695$
1	1	0.15	0.5374	0.4626	$b_{new} = 0.05 + 0.01(0.3877)$
0	0	0.05	0.5125	-0.5125	$b_{new} = 0.088$
0	0	0.05	0.5125	0.4875	$w_1 = -0.005$
1	0	0.05	0.5125	-0.5125	$w_2 = 0.1 + 0.09252 \Rightarrow 0.192$

Iteration 3: $\alpha = 0.1$, $w_1 = 0.005$, $w_2 = 0.192$, $b = 0.088$

W	H	z	P	$e = y - p$	Loss = $(-1/5)(-3.252)$
0	1	0.28	0.57	0.43	$\Rightarrow 0.65$
1	1	0.275	0.568	0.432	$b_{new} = 0.088 + 0.1(0.295)$
0	0	0.088	0.523	-0.523	$\Rightarrow 0.118$
0	0	0.088	0.523	0.477	$w_1 = -0.005 + 0.1(-0.089)$
1	0	0.083	0.521	-0.521	$\Rightarrow -0.014$

$$w_2 = 0.192 + 0.1(0.862) \Rightarrow 0.2782$$

$$b = 0.118, \quad w_{12} = 0.014, \quad w_2 = 0.2782$$

w_1	H	z	$P(\text{Rain})$	Prediction	Action
0	1	0.3962	0.593	Rain	Do No Delivery
1	1	0.3822	0.594	Rain	No Delivery
0	0	0.118	0.529	Rain	No Delivery
0	0	0.118	0.529	Rain	No Delivery
1	0	0.104	0.523	Rain	No Delivery

4. Comparison:

Naive Bayes	Entropy	Logistic Regression
Outputs Probability	Outputs Class	Outputs Probability
Learns Probabilities given a class	Learns Rules	Learns weights and bias
Handles small data well	Overtits on small data	Needs regularization on small data
Training Acc: 80%	Training Acc: 80%	Training Acc: 60% ↓ Based on 3 iterations

Q5.

1. Joint Probability Distribution:

$$P(\text{Infected}) = P(\text{Healthy}) \Rightarrow 0.5$$

	A	B	Probability
Infected	Infected	Infected	0.25
Infected	Healthy	Infected	0.25
Healthy	Infected	Healthy	0.25
Healthy	Healthy	Healthy	0.25

2. Action Table:

Timestamp	Drone Pos	A	B	Percept	Action	Credit
$t=0$	A	Infected	Infected	[A, Infected]	Spray	0
$t=1$	A	Healthy	Infected	[A, Healthy]	Move to B	\$2
$t=2$	B	Healthy	Infected	[B, Infected]	Spray	\$2
$t=3$	B	Healthy	Healthy	[B, Healthy]	Move to A	\$4

Total credits = \$8

Q6.

A). FAIRGAME: a Framework of AI Agents Bias Recognition using Game Theory

1. What is the role of Game theory in AI Agent collaboration?

Game theory formalizes multi-agent interactions as games to model, predict and optimize behavior of rational agents.

2. What is the importance of game theory?

- Model interactions between agents

- Allows measuring bias detection, interpretability, and trust in multi-agent systems.

- Provides benchmarks for comparing observed AI behavior with theory.

3. What are the two classical Game theories presented in the paper? And how it can be used for 5G applications?

1. Prisoner's Dilemma: Players gain by defecting or cooperating with mutual cooperation leading to better payoffs.

2. Battle of the Sexes: A coordination game where players prefer different outcomes but value coordinating.

Prisoners Dilemma can model cooperation between base stations for spectrum sharing and interference management, while Battle of the Sexes can model coordination in network slicing, handovers, and edge-computing resource allocation.

4. Enlist selected LLMs presented in the paper.

Meta: Llama 3.1 (405b)

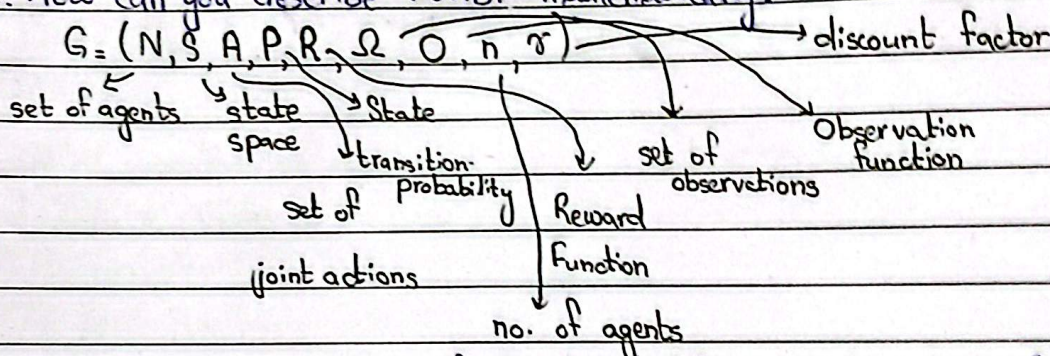
Mistral AI: Mistral Large (123b)

OpenAI: GPT-4

Anthropic: Claude 3.5 Sonnet

B). A semi-independent policies training method with shared representation for heterogeneous multi-agents reinforcement learning

1. How can you describe POMDP mathematically?



2. What are the application of Hard and Soft parameter sharing?

① Hard Parameter Sharing:

- One shared network learns common features for multiple related tasks.
- Can be used in Multi-Agent Reinforcement Learning so agents can learn from each other's experiences.
- Situations where resource limitation is requirement.

2. Soft Parameter Sharing:

- Each task maintains its own model and parameters.
- Useful when independent agents require independence.
- For tasks having different input and output requirements.

3. Explain and decode equation no 1

$$L_e(\theta_e) = \mathbb{E}_{(o, a, r, o') \sim D} \left[\sum_i \|f_o(z_{a,i}, o_i, a_{-i}) - o'_i\|_2^2 + \lambda_e \sum_i (f_r(z_{a,i}, o_i, a_{-i}) - r_i)^2 \right]$$

This equation defines the loss function used to train the encoder-decoder prediction model.

$\mathbb{E}$ → expected loss averaged over samples from the replay buffer. D → Shared replay buffer containing agents' experiences.

o'_i → Actual next observation of agent 'i'

$f_o(z_{a,i}, o_i, a_{-i}) = \hat{o}_i$ → Predicted next observation of agent 'i'

r_i → Actual reward received by agent 'i'

$f_r(z_{a,i}, o_i, a_{-i}) = \hat{r}_i$ → Predicted reward of agent 'i'

λ_e → Weight which controls the importance of reward-prediction error

$$\sum_i \| \hat{o}_i - o_i \|_2^2$$

This is Observation Prediction Loss and it measures the difference between the next observation by the model and the observation actually received by each agent.

$$\lambda_e \sum_i (\hat{r}_i - r_i)^2$$

This is Reward Prediction Loss and it measures the difference between the predicted and actual reward. The λ_e ~~part~~ controls the contribution of this error to the total loss.

4. Explain intuition behind figure no 4 presented in the paper.

The figure illustrates how the proposed method combines two intrinsic rewards to train heterogeneous multi-agent reinforcement learning agents:

1. Exploration reward: Encourages agents to explore unfamiliar states and learn more about the environment.
2. Representation Diversity Reward: Encourages agents of same type to learn similar representations, while other agents learn distinct representations.

These intrinsic rewards are combined with environment's original reward to guide learning. The goal is to encourage ~~max~~ exploration while helping agents learn specialized policies according to their types.

Q7.

$$1. A = (6, -1, \sqrt{9}), B = (4, x, -2)$$

$$A \cdot B = 0$$

$$(6)(4) + (-1)(x) + (3)(-2) = 0$$

$$18 - x = 0$$

$$x = 18$$

2. Dot product between vectors results in a scalar, meanwhile, ~~Ham~~ Hadamard product is a vector.

3. From figure:

$$u = (1, 2), v = (2, 2)$$

$$u \cdot v = \|v\| \cdot \|u - \text{proj}\|$$

$$\|v\| = \sqrt{2^2 + 2^2} \Rightarrow 2\sqrt{2}$$

$$\|u - \text{proj}\| = \sqrt{1^2 + 1^2} \Rightarrow \sqrt{2}$$

$$u \cdot v = 2\sqrt{2} \times \sqrt{2} \Rightarrow 4$$

$$(1, 2) \cdot (2, 2)$$

$$\Rightarrow 2 + 4 = 6$$

Hence Proved!

$$Q8. \text{Agent } A = 0.2t + t^2$$

$$\text{Agent } B = \sin(t) + \cos(t)$$

$$\text{Agent } C = t^5 + t^6 + \sin(t)$$

$$\text{Agent } D = t^5 + t^8 + \sin(t) + \cos(t)$$

$$S(t) = \text{Agent } A(t) + \text{Agent } B(t) + \text{Agent } C(t) + \text{Agent } D(t)$$

$t = 1:$

$$A = 0.2(1) + (1)^2 \Rightarrow 1.2, B = \sin(1) + \cos(1) \Rightarrow 1.017$$

$$C = (1)^5 + (1)^6 + \sin(1) \Rightarrow 2.017, D = (1)^5 + (1)^8 + \sin(1) + \cos(1) \Rightarrow 3.017$$

$t = 2:$

$$A = 0.2(2) + (2)^2 \Rightarrow 4.4, B = \sin(2) + \cos(2) \Rightarrow 1.034$$

$$C = (2)^5 + (2)^6 + \sin(2) \Rightarrow 96.034, D = (2)^5 + (2)^8 + \sin(2) + \cos(2) \Rightarrow 289.03$$

$t = 3:$

$$A = (0.2)(3) + (3)^2 \Rightarrow 9.6, B = \sin(3) + \cos(3) \Rightarrow 1.051$$

$$C = (3)^5 + (3)^6 + \sin(3) \Rightarrow 972.052, D = 6805.051$$

$$t = 4:$$

$$A = 0.2(4) + (4)^2 \Rightarrow 16.8, \quad B = 1.067$$

$$C = (4)^5 + (4)^6 + \sin(4) \Rightarrow 5120.07, \quad D = (4)^5 + (4)^6 + \sin(4) + \cos(4) \Rightarrow 66561.067$$

$$S(1) = 1.2 + 1.017 + 2.017 + 3.017 \Rightarrow 7.251$$

$$S(2) = 4.4 + 1.034 + 96.034 + 289.034 \Rightarrow 390.502$$

$$S(3) = 9.6 + 1.051 + 972.052 + 6805.051 \Rightarrow 7787.754$$

$$S(4) = 16.8 + 1.067 + 5120.07 + 66561.067 \Rightarrow 71789.004$$

The distribution is polynomial as the term t^6 dominates as time passes.

